PAGE NO. :
DATE : / /

Q.3 Objective : $J(\theta) = \frac{1}{2} \| X\bar{\theta} + X_i \theta_i - \vec{y} \|_2^2 + \lambda \|\bar{\theta}\|_1$

$+ \lambda s_i \theta_i$

Fixing the sign $s_i \in \{-1, 1\}$

$$J(\theta) = \frac{1}{2} \| X\bar{\theta} + X_i \theta_i - y \|_2^2 + \lambda s_i \theta_i$$

Differentiate w.r.t θ_i

$$\frac{dJ}{d\theta_i} = X_i^T (X\bar{\theta} + X_i \theta_i - y) + \lambda s_i$$

Setting derivative to 0, $0 = X_i^T (X\bar{\theta} + X_i \theta_i - y) + \lambda s_i$

Let $\rho = X_i^T (y - X\bar{\theta})$

So, $X_i^T X_i \cdot \theta_i = \rho - \lambda s_i$

$$\theta_i = \frac{\rho - \lambda s_i}{X_i^T X_i}$$

Combining both cases,

$$\theta_i = \begin{cases} \frac{\rho - \lambda}{X_i^T X_i} & \text{if } \rho > \lambda \\ \frac{\rho + \lambda}{X_i^T X_i} & \text{if } \rho < -\lambda \\ 0 & \text{if } \rho \leq \lambda \end{cases}$$

This is the soft-thresholding update which drives small coefficients zero.